



Artificial Intelligence and Machine Learning

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AI Landscape

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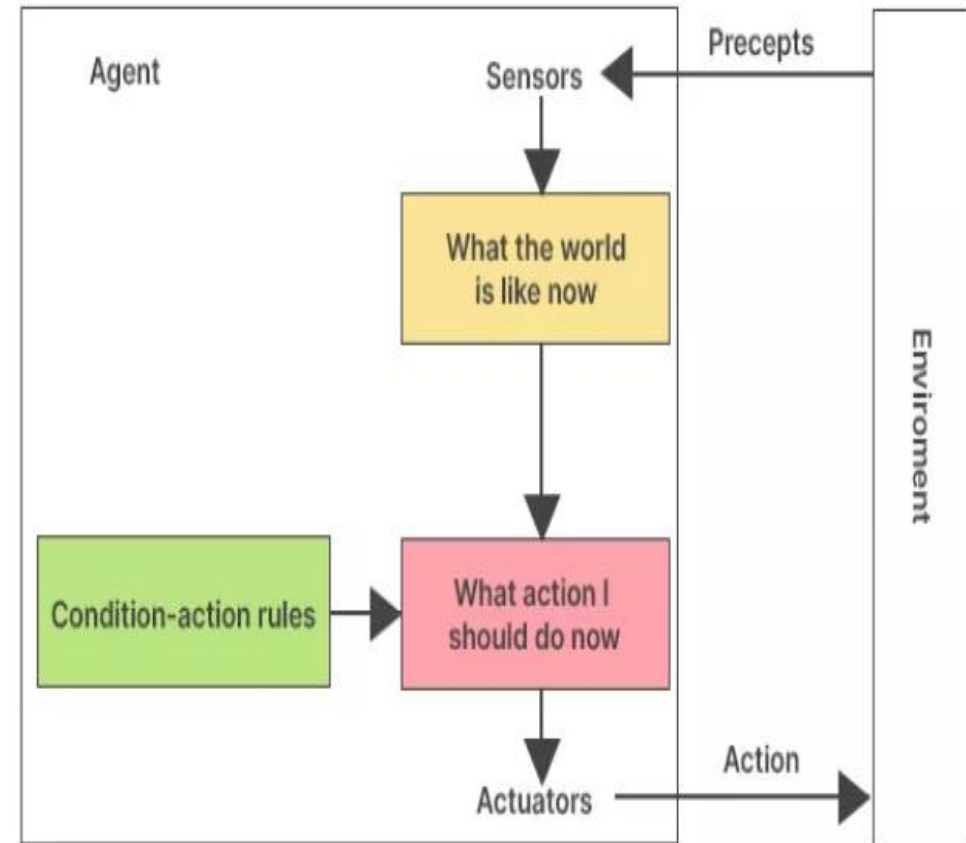
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What is an Environment?

The Environment is everything the agent interacts with.

It provides percepts to the agent and is affected by the agent's actions.

Example: For a vacuum robot, the environment is the floor layout, obstacles, and dust



Environment Types

1. Fully or Partially Observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continuous
6. Single-Agent vs Multi-Agent

Fully or Partially Observable

Fully Observable

- Agent has access to the complete state of the environment at each moment.
- Example: Chess (all pieces visible)

Partially Observable

- Agent has limited access to environment information
- Example: Driving in fog, Poker

Deterministic Vs Stochastic

Deterministic

- Next state is completely determined by the current state and action.
- Example: Crossword puzzle

Stochastic

- Includes randomness and uncertainty.
- Example: Rolling a dice, stock market

Episodic Vs Sequential

Episodic

- Each action is independent of previous actions
- Example: Image classification

Sequential

- Current decisions affect future actions
- Example: Chess, Driving

Single-Agent Vs Multi-Agent

Single-Agent

- Agent operates alone
- Example: Puzzle-solving bot

Multi-Agent

- Multiple agents interact or compete
- Example: Game of soccer, autonomous cars on road

Example: Environment of Google Map Agent

Partially observable (can't know all traffic details)

Stochastic (traffic may change)

Sequential (each route decision affects others)

Dynamic and Multi-agent (other drivers)

Performance Measure: Time taken, route accuracy, fuel efficiency

“Environment affects how the agent behaves”

How do we design an agent properly for a given environment?

“For building an agent, define:

- What success means
- Where it operates
- What it can sense
- What it can do

PEAS Model

PEAS Model

It is a Framework for Evaluating and Designing Rational Agents

P: Performance Measure

E: Environment

A: Actuators

S: Sensors

Performance Measure (P)

This defines how the agent's success is measured, such as the number of tasks completed, the time taken, or the accuracy of its actions.

Environment (E)

This refers to the surroundings in which the agent operates, including its physical or virtual context.

Actuators (A)

These are the mechanisms through which the agent interacts with and changes its environment, such as motors, speakers, or display screens

Sensors (S)

These are the tools used by the agent to perceive and gather information about its environment, such as cameras, microphones, or touch sensors

PEAS	Derive Answers for Questions
Performance	How do we measure success?
Environment	Where does agent operate?
Sensors	How does it perceive?
Actuators	How does it act?

PEAS Model for Driverless Cars

Performance Measure: The measure for driverless cars is safe navigation and efficient route planning, ensuring passenger safety and timely arrivals.

Environment: The environment includes roads, traffic patterns, pedestrians, and weather conditions, which the car must interact with while navigating.

Actuators: Actuators consist of steering, acceleration, and braking systems that execute the car's movements as directed by its AI algorithms.

Sensors: Sensors, such as cameras, LiDAR, GPS, and radar, collect real-time data about the car's surroundings



THANK YOU

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